The understanding on Fedora and its application on Memorial Sloan-Kettering Cancer Center data system

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Fedora is an entirely new system totally different from the traditional RDBMS one. However, there are still some similar places between them. The first, their processing target are objects. The second, they likewise employ relationship as a tool to connect objects. The third, they concurrently attempt to handle the real world objects and improve the efficiency of information retrieval.

Nevertheless, there are some big differences between Fedora and traditional RDBMS.

On the aspects of object handling, Fedora attempt to abstract anything from real world into object, and seek to provide a series of method to manipulate them. RDBMS mainly employ data as intermediary to handle the objects with correspondence to the one in real world. Particularly, Fedora places its emphasis on documents, image, video, audio, archives, and collections, so, metadata naturally become a part of object. However, to be different with Fedora, RDBMS places its initial emphasis or highlight on data. When dealing with sophisticated objects hard to disintegrate, such as video, audios, and archives, it, then, on the one hand store them into database; on the other hand extract metadata from them as a representation of them. From this vision, Fedora is more suitable to handle complicated objects than RDBMS.

On the object scope aspects, Fedora is mainly used to handle the objects existed in whatever any place as long as it can reach them through link. RDBMS is restricted to a relatively enclosed environment and have a relatively weak power in handling the objects out of database scope. For example, although SQL language under RDBMS, such as Oracle, can store the large XML files as an objects, it can not directly retrieve the data contained in this XML files. It must firstly disintegrate and extract data contained in XML, store them, and then retrieve them. In this aspect, fedora is perfect because it adopt a strategy to embed parsing tools into system and automatically retrieve the data contained in XML.

On the container layer, Fedora mainly uses XML as the media or agent to store most of the objects, and related metadata, controlling information, such as persistent ID, object properties, and contents, such as related datastream. However, RDBMS mainly uses database system to be an infrastructure to support related functions, and it put everything, whatever objects or metadata into tables or databases. Here, it is hard to say which method is better on handling objects. The only thing we can conclude is that they provide services with different target and different emphasis. For example, RDBMS mainly faces to business and industry world, with restrict requirements on data quality; and under rigid or inelastic environment, it does not allow any errors happened in most situation. However, Fedora faces a knowledge transferring context. So, its main target is not to ensure 100 percent accuracy or presicion. Its target is mainly to interpret and analyze the complicated relationship among numeric objects, and demonstrate them to the people who want to know them. Hence, Fedora and RDBMS have no collisions.

On the server technique aspect, Fedora mainly uses or employs content server technology, which is totally different with the server technology used by RDBMS. Fedora actually employs a virtual server technique to ensure it can go through the entire relationships built upon objects to retrieve the useful objects, and represent them to users. Because Fedora is a high level concept, we can use it without considering or worrying about how to realize or accomplish bottom layer techniques. However, RDBMS, compared with Fedora, must put much more energy to consider deep layer design to ensure its operation in high efficiency. From this aspect, Fedora is relatively easy to implement compared with RDBMS system.

On the model building aspects, Fedora mainly uses or employs content model to simulate object relationships in the real world, to ensure delivering “essential characteristics” of objects smoothly. However, RDBMS use entity-relationship(ER) model to simulate the real world objects relationship. According to the illustration of Fedora, it can handle the complicated relationship among millions of objects. However, if using this way, RDBMS perhaps become crashed for the reason of its technical limit on handling isolated objects relationship. But, RDBMS can employ another strategy to solve this problem. It can classify and catalog the similar objects into cluster, and store them into classified tables, then handle the relationship among them. In this way, RDBMS can also handle relationship among millions of objects. However, RDBMS does not provide convenient tools, with automatic parsing ability, to classify and cataloging large number of heterogeneous objects. From this aspect, Fedora is obviously more competent than RDBMS.

On the aspect of maturity, for the reason of XML still in development (XML version 1.0), and the complication of relationship in real world, Fedora still require lots of endeavor to push its concept into reality. However, RDBMS is relatively mature compared with Fedora after so many years development since Codd brings forward famous 12 rules. But, for the circumscribed characteristics of RDBMS, although it is mature, it is not suitable to handle large number of sophisticated heterogeneous objects.

On the artificial intelligence aspects, Fedora still requires combining semantic research theory into its platform or product, to ensure the accurate relationship built among objects. In some degree, Fedora is in the front end of artificial intelligence, because merely depending person to recalculate the relationship among millions of objects according to context, is actually impossible. In opposition, RDBMS, at earth, seldom put emphasis on how to build relationships automatically by use of artificial intelligence combination way when handling objects. If required, RDBMS undoubtedly requests engineer to involve and setup object relationships through use of ERD by hand, step by step. On this aspect, Fedora has greater and more perfect perspective than RDBMS.

On the running speed, Fedora is restricted by some bottom layer technique, such as java running efficiency, server speed, and algorithms optimization. Some semantic analysis do consume most of the CPU time, so algorithms determine whether Fedora is competent on the process of recalculating relationships among objects. RDBMS mostly running on the database platform, with backbone software compiled by C++, so its running speed is much more faster. However, in some situation, because Fedora and RDBMS deal with or cope with different situations and tasks, it is hard to compare between them.

Under the concrete situation of MSK, its library mainly provides services for three kinds of target users, who separately are researchers or medical staffs in MSK, students, administrative staff, and patients/social users. Simultaneously, there are many kinds of heterogeneous resources hold in library repository. Hence, from my understanding, the main task of MSK library is to provide library information services to target user groups with customization and in high quality. This target decides which kinds of web page layout should take, which kinds of strategy should adopt to satisfy the user or patron requirements.

From analysis, the most important service target for library should be the medical staff, researchers in MSK, and some students related with MSK. The second important one should be administrative staff in MSK. The last one should be the patients and other social users. Under this context, the database, E-journals, E-books, and internet resources, on

either clinical aspects, or science research aspects, should be the first emphasis MSK library should place. The resources facing administrative staff should be in the second layer emphasis of library. The cancer information facing patients and common users should be in the third position of library work.

Although MSK library provides the resources catalog for clinical services/research and basic science research staff or programs, these catalog are just comprehensive ones which merely tell the researchers of what kinds of database possibly contain what aspect of information and it can not control the delicate contents presentation. The best way to utilize and represent the information contained in these database and E-journals is to build a schema to extract special information from them to match the related catalog or staff query request. The administrative staff maybe has the same requirements on data query and utilization. The key problem, here, is that it definitely needs a complicated metadata system, extracted from resources, to match the researchers or staff query request, not only in high recall, but also in high precision. However, it is not easy to reach this target if without a powerful tool as a help.

As for the special context MSK library stands in, there are two kinds of way to handle this situation.

The first, it is definitely suitable to use the Fedora system or concepts to build this metadata system, and put articles contained in journals or databases as the objects, simultaneously adding digital object identifier, object properties, and datastreams on them, forming or shaping a satisfying Fedora object collections. After completing this, some XML handling language, such as XQuery, and some XML demonstration language, such as XSLT, can be employed to search object and represent them to the researchers.

The second, it can use information retrieval software, such as Lucene, to extract token words from these documents, and built a large index library based on related articles collection. Then, when user or patron searches by inputting keywords or natural language, Lucene system employ algorithms to judge what kinds of documents and how many of them match the user query, and return the related documents according to relevance degree calculated.

The two kinds of method above concurrently have advantage and disadvantage. The decision in choosing any one of them greatly depends on the technical strength the MSK library holds, and deep understanding on MSK context. In normal situation, if technical strength and support is strong enough, Fedora is a good choice.